ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 5

Spatial Relationships

Week 3 | L5

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Everything is related to everything else, but near things are more related than distant things.

— Waldo Tobler, 1970

This single principle underpins almost all spatial analysis:

Nearby soil samples have similar properties

Adjacent buildings share earthquake damage patterns

Water quality at one point predicts quality nearby

Traffic at one intersection affects nearby intersections

If this law didn't hold, spatial analysis would be pointless -- we couldn't predict anything from location.

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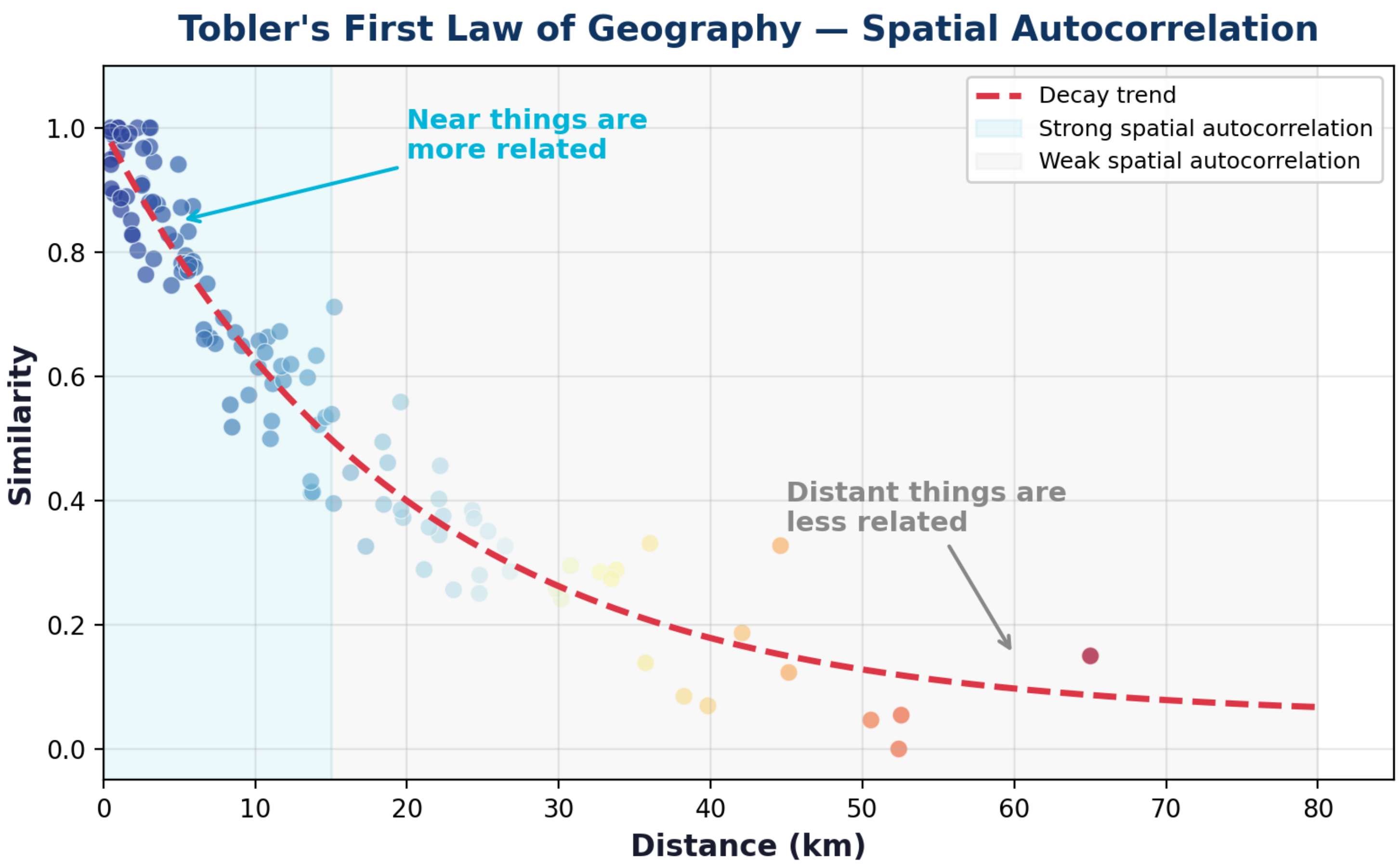
Spatial Autocorrelation



"Everything is related to everything else, but near things are more related than distant things." — Waldo Tobler, 1970

Positive autocorrelation: similar values cluster (most common). Negative: dissimilar values cluster. Zero: no spatial pattern.

Spatial Correlation Across Layers — a 100-Million-Year Example

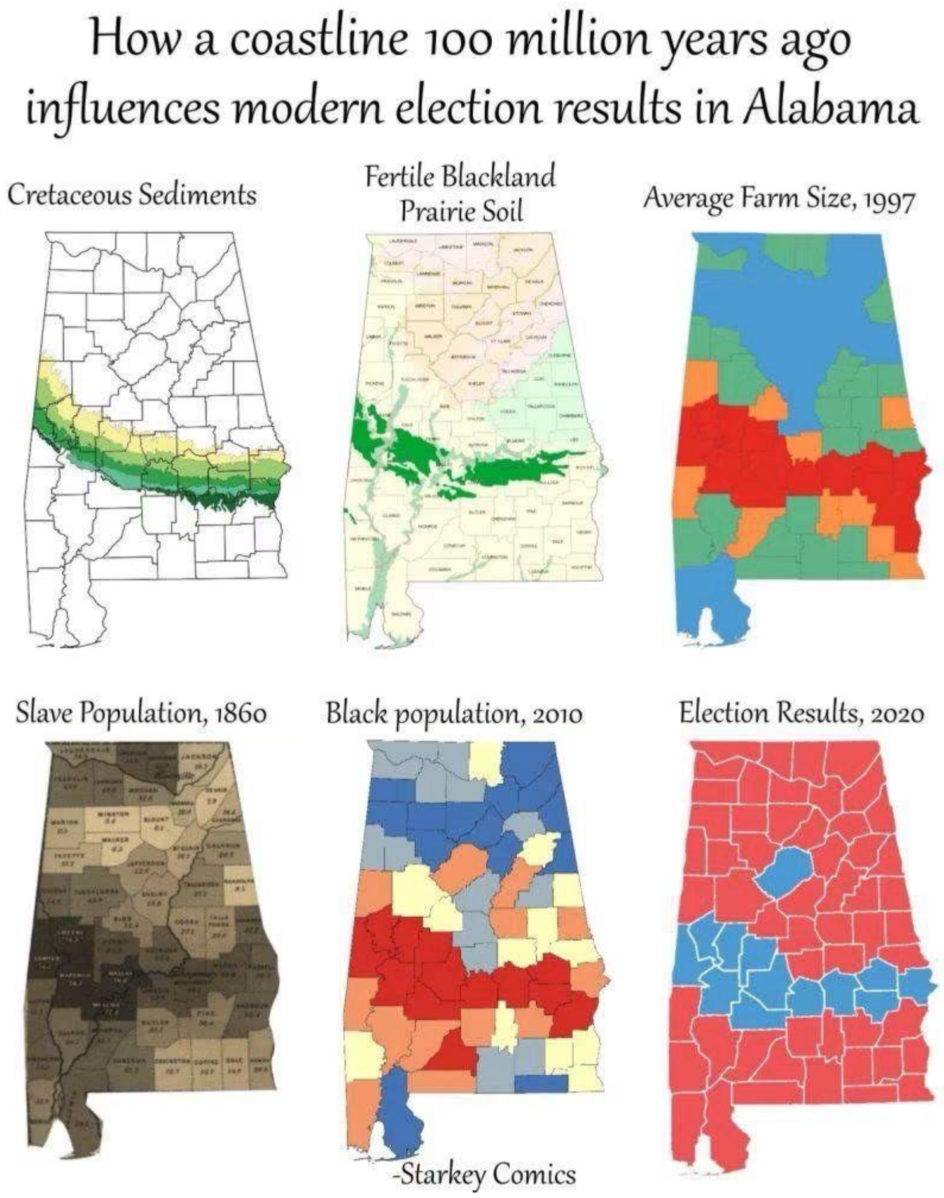
Six layers of Alabama data — geology, soil, farming, demographics, politics — all show the same crescent pattern.

Cretaceous chalk → fertile 'Blackland Prairie' soil → cotton plantations → 1860 enslaved population → 2010 Black population → 2020 election results

The pattern in one layer PREDICTS the pattern in the others — that is spatial correlation between layers

Overlaying layers reveals the mechanism; a scatter plot of the same data never could

Caution: correlated maps show association, not proof of cause — the causal chain must be argued, not just mapped.



Source: Starkey Comics

Four Fundamental Spatial Relationships

The building blocks of spatial analysis in GIS.

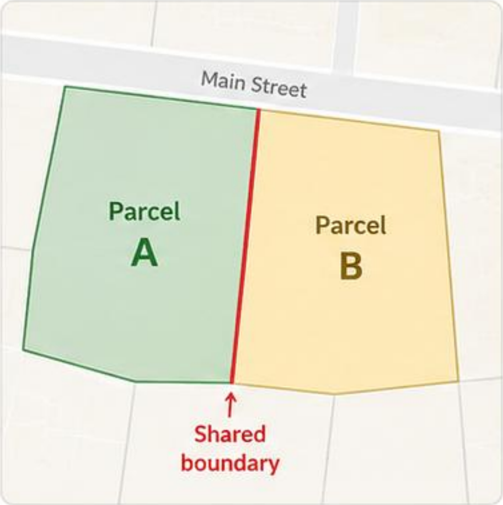
1. Proximity

Objects are near each other in space.

GIS Question:
How far apart?

2. Adjacency

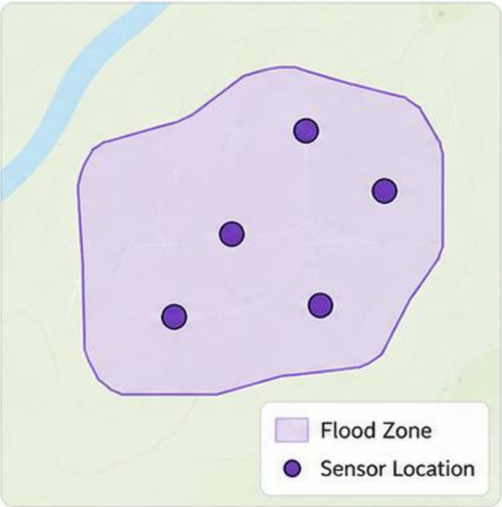
Objects share a common boundary or are next to each other.



GIS Question:
Do they share a boundary?

3. Containment

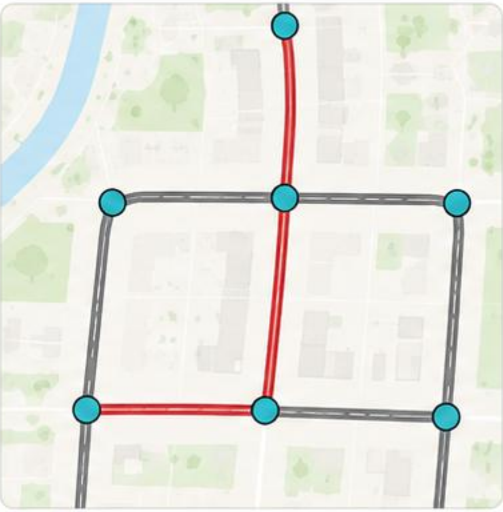
One object is completely inside another.



GIS Question:
Is this inside that?

4. Connectivity

Objects are linked or connected by a path or network.



GIS Question:
Is there a path between?

Why it matters: These relationships help us answer real-world questions about location, patterns, and connections.

Every spatial analysis question boils down to one (or more) of these four relationships.

The tools change (buffer, join, overlay) -- but the concepts stay the same.

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Proximity -- How Far Apart?

Distance is the foundation of spatial analysis

Proximity questions in engineering:

How far is each building from the nearest fire station?

Which pipes are within 100m of a known contamination source?

What is the average spacing between boreholes in this survey?

Tools for proximity:

Buffer -- create a zone around features at a specified distance

Nearest neighbour -- find the closest feature

Distance matrix -- calculate all pairwise distances

Remember: distance must be calculated in a projected CRS (metres), not degrees!

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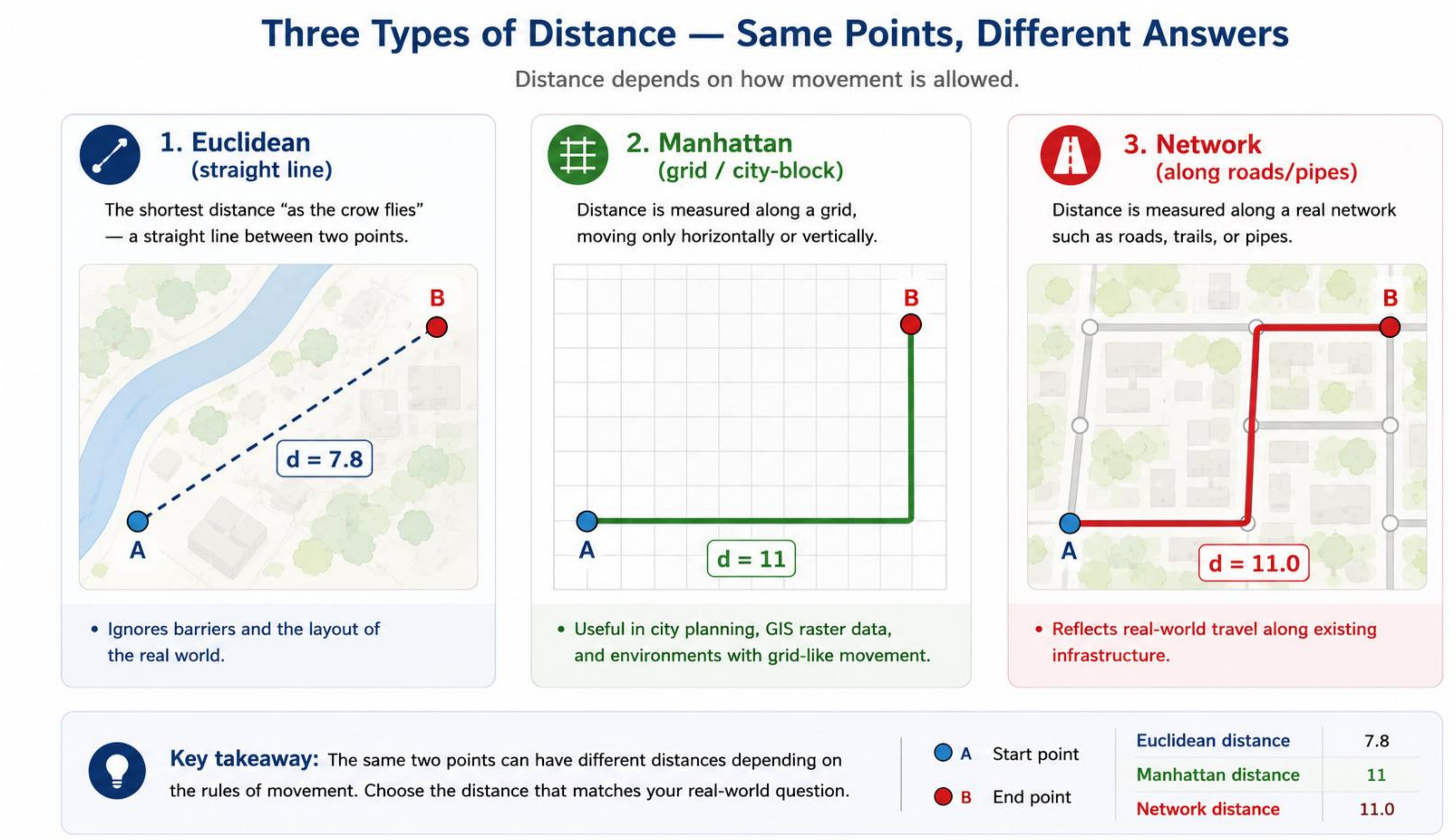

The "right" distance depends on the question: straight line for air quality, network for emergency response.

Always ask: 'straight-line or network distance?'

Adjacency & Topology -- Shared Boundaries Matter

Adjacency asks: do these features share a boundary?

Engineering examples:

Connected pipe segments — flow depends on connections

Neighbouring land parcels — planning rules apply to neighbours

Adjacent mesh elements — load transfer in FEM

Shared catchment boundaries — water flows across them

Topology:

The study of spatial relationships that are preserved under deformation

Properties that survive stretching:

Connectedness (is there a path?)

Adjacency (do they touch?)

Containment (is it inside?)

Properties that DON'T survive:

Distance, angle, area — these need metric CRS

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Containment -- Is This Inside That?

Point-in-polygon is the classic containment query

Engineering examples:

Which buildings are inside the flood zone?

Which boreholes are within the project boundary?

Which pipes pass through unstable soil?

Which road segments are within the earthquake exclusion zone?

In Python -- Spatial Join:

```
result = gpd.sjoin(buildings, flood_zone, predicate='within')
```

This returns all buildings that are WITHIN the flood zone, with attributes from both layers.

Spatial join = the most powerful single operation in GIS.

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Connectivity -- Network Relationships

Can you get from A to B?

Network analysis asks:

Is the pipe network connected? (no isolated segments)

What is the shortest path from A to B?

If this road is blocked, is there an alternative route?

How many people can be served from this water source?

Engineering networks:

Road networks -- routing, accessibility

Pipe networks -- flow, capacity, redundancy

Power grids -- supply, failure cascades

River networks -- drainage, catchments

Connectivity is topological — it depends on how things are linked, not on exact positions or distances.

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Topological Predicates -- The Full Set

Predicates (True/False tests):

intersects — any shared space?

within — entirely inside?

contains — contains entirely?

touches — shared boundary only, no interior overlap?

crosses — passes through?

overlaps — partial shared interior?

disjoint — no shared space at all?

Most commonly used:

intersects -- the most permissive

(True if geometries share ANY space)

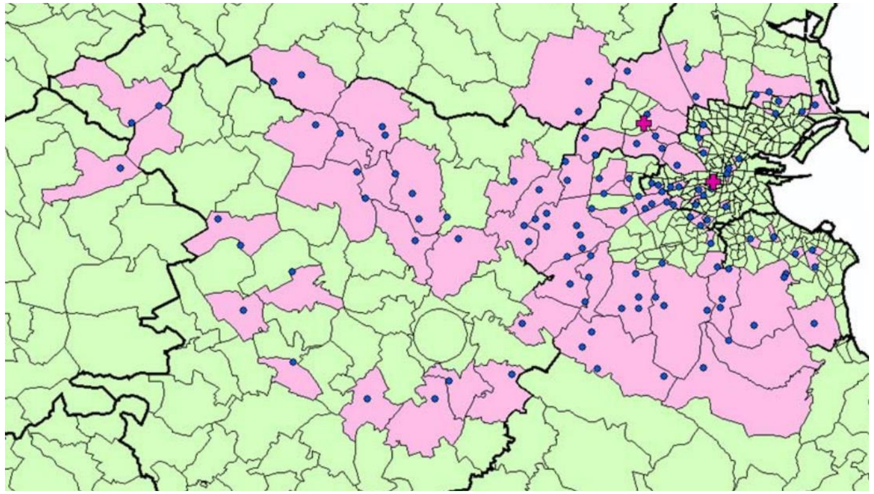
within -- the most restrictive

(True ONLY if entirely inside)

In GeoPandas:

gdf1.intersects(gdf2)

gpd.sjoin(gdf1, gdf2, predicate='within')



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LIVE DEMO

L05_SpatialQueries.html

Explore spatial predicates visually — drag geometries and see which predicates are True/False in real time.

Scan to open the demo on your phone

aardid.github.io/uc_205_gis_course/L05_SpatialQueries.html

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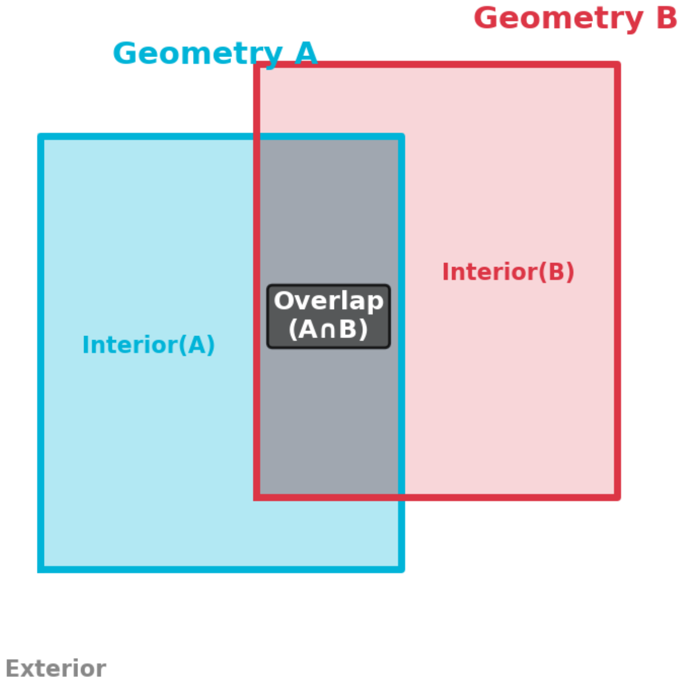
DE-9IM -- The Mathematics Behind Predicates

DE-9IM — The Mathematical Basis for Spatial Predicates

Overlapping Geometries



DE-9IM Matrix

	Interior(B)	Boundary(B)	Exterior(B)
Interior(A)	2	1	2
Boundary(A)	1	1	1
Exterior(A)	2	1	2

2 = area | 1 = line | 0 = point | F = empty

DE-9IM (Dimensionally Extended 9-Intersection Model) defines ALL spatial predicates mathematically. You don't need to memorise it -- just know it exists.

Spatial Predicates in Python

Boolean tests (True/False for each row):

```
gdf1.intersects(gdf2)     # Series of True/False
gdf1.within(gdf2)         # is each feature of gdf1 within gdf2?
gdf1.contains(gdf2)       # does each feature of gdf1 contain gdf2?
gdf1.touches(gdf2)        # do they share only a boundary?
gdf1.disjoint(gdf2)       # no shared space at all?
```

Spatial join (combine attributes based on location):

```
result = gpd.sjoin(left_gdf, right_gdf, how='inner', predicate='intersects')
```

Filtering (select features that meet a spatial condition):

```
flood_buildings = buildings[buildings.intersects(flood_zone.union_all())]
```

These three patterns cover 90% of spatial analysis tasks.

Always ask: 'which relationship do I actually mean?'

When Tobler's Law Fails

Tobler's law is a generality -- not a universal truth

Abrupt boundaries:

Geological faults — properties change dramatically across a fault line

Zoning boundaries — land value jumps at a zone boundary

River banks — flood risk is high on one side, low on the other

Infrastructure networks:

Two houses 50m apart but on different water supply networks

A building next to a road vs across a motorway — close but disconnected

Long-range connections:

Pollution from a distant factory carried by wind or water

Earthquake waves affecting distant structures on the same soil type

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KEY CONCEPTS

Today's Takeaways

1. Tobler's First Law: near things are more related -- the basis of all spatial analysis
2. Four spatial relationships: proximity, adjacency, containment, connectivity
3. Distance type matters: Euclidean vs Manhattan vs network-- choose based on the question
4. Spatial predicates (intersects, within, touches, etc.) are Boolean tests for relationships
5. Spatial join is the most powerful GIS operation -- combining data by location

NEXT LECTURE

Lecture 6: Vector Analysis Operations

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Spatial Dependence in Engineering

Spatial dependence = observations at nearby locations are not independent

Why this matters:

Groundwater levels — wells 100m apart give similar readings (not independent samples)

Earthquake damage — adjacent buildings share the same ground conditions

Soil properties — closely spaced boreholes reveal similar soil types

Air quality — neighbouring sensors measure similar pollution levels

Implications:

You can interpolate between measurements (kriging, IDW)

But nearby samples add less information than distant ones

Standard statistics assumes independence — spatial data violates this!

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For a pipe network analysis, which spatial relationships matter most?

Think about:

Do you need proximity (how far from buildings), adjacency (connected segments)?

Do you need containment (which pipes are in which soil zone)?

Do you need connectivity (can water flow from source to tap)?

Which type of distance matters — Euclidean or network?

Engineering Example: Buildings Near a Fault Line

Question: Which buildings are within 500m of the Greendale Fault?

Step 1: Load building footprints and fault line data

Step 2: Reproject both to EPSG:2193 (NZTM — metres)

Step 3: Buffer the fault line by 500m

```
fault_buffer = fault.buffer(500)
```

Step 4: Spatial join — which buildings intersect the buffer?

```
at_risk = gpd.sjoin(buildings, fault_buffer, predicate='intersects')
```

Step 5: Count and report — e.g. 342 buildings within 500m

This 5-step pattern (load → reproject → buffer → join → report) solves most proximity+containment questions.

Always ask: 'did I reproject first?'

From Relationships to Operations -- Preview of Lecture 6

Spatial relationships tell us WHAT to ask.

Spatial operations tell us HOW to answer.

Relationship → Operation:

Proximity → Buffer, Nearest Neighbour

Containment → Spatial Join, Clip

Overlap → Overlay (union, intersection, difference)

Adjacency → Dissolve, Spatial Weights

Lecture 6 will cover:

Buffer — create proximity zones

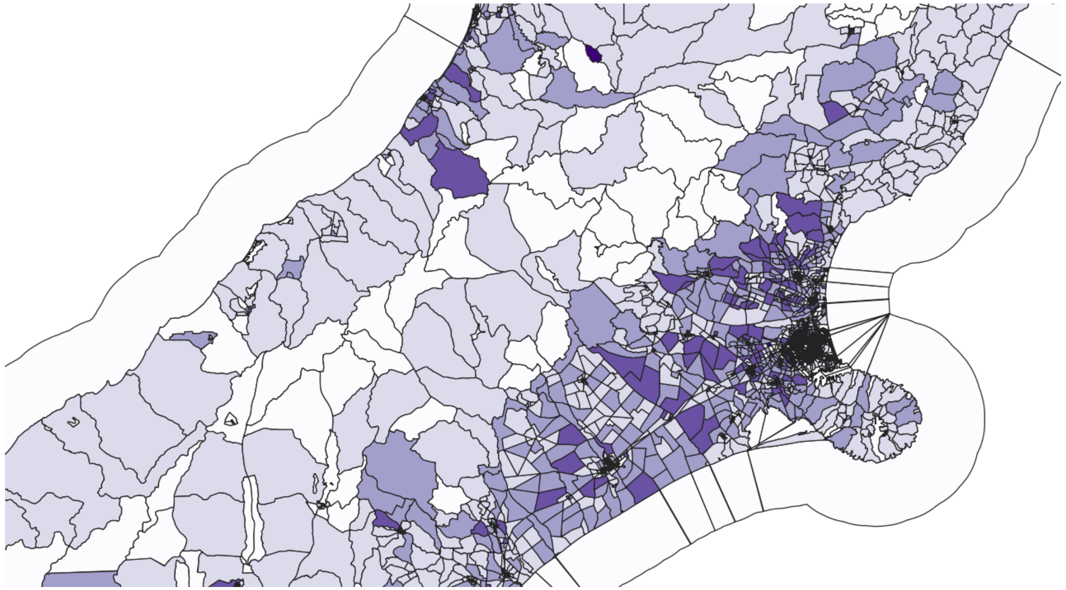
Clip — cut data to a boundary

Spatial Join — combine by location

Overlay — union, intersection, difference

Dissolve — merge features by attribute

Today: the concepts. Next lecture: the tools.



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